

CNS summary for Abstract:

A Proposed Communication, Navigation & Surveillance System Architecture to Support Urban Air Traffic Management

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- The key technologies enabling UATM are Communications, Navigation and Surveillance (CNS) systems. This paper presents a proposed configuration for CNS System to support UATM. Aircraft for UAM are supposed to operate within the aerial bounds of the cities away from conventional air traffic. Thus air traffic would be much denser and the Air Traffic Management (ATM) system required to control air mobility must provide integrity, robustness, security and high geo-spatial accuracy. Additionally, there is also a need for fast and accurate communication between traffic controller and the air vehicle. Communication requirements for UAM include Vehicle-to-Vehicle (V2V), Vertiport to vertiport and Vertiport to Vehicle communication. Navigation requires Self-positioning and Situational awareness. Surveillance needs to be able to perform Ground stations detecting and tracking aircraft, position and Identification updates. The paper presents the concept of operation, architecture of CNS systems fulfilling these requirements.
- Choice of CNS based on Concept of Operation
- Multipath are considered the dominant error sources in urban settings.
- 5G network that has already been proposed as back up for communication can provide node-to-node communication capabilities that lead to real time channel estimation and thus alternate navigation techniques [3].

ADVANCED AIR MOBILITY Research Directions for Communications, Navigation, and Surveillance Kamesh Namuduri , Uwe-Carsten Fiebig , David W. Matolak , Ismail Guvenc , K.V.S. Hari, and Helka-Liina Määtänen

- Industry is moving fast with excitement to deploy AAM solutions. However, there are multiple technical challenges that need to be overcome before AAM becomes a reality. This article takes a closer look at the technology readiness level of AAM solutions in the area of communications, navigation, and surveillance (CNS) and identifies open research problems as well as directions to address them.
- While a UTM is designed for small UASs flying at or below 400 ft above ground level (AGL), AAM includes larger aircraft carrying people and/or cargo at altitudes between 500 and 2,000 ft AGL. AAM platforms include newer aircraft designs with vertical takeoff and landing capabilities.
- A key subsystem in the AAM architecture [2] is the provider of services for UAM (PSU), which will provide services to support operations planning, flight intent sharing, strategic and tactical deconfliction, and airspace management functions. PSUs will exchange information with other PSUs via a network that enables safe, efficient operation within the AAM corridors—volumes of airspace dedicated to AAM use.
- ASTM International are being followed for data exchange protocols between one subsystem to another within the AAM system.
- this article highlights some of the most critical CNS challenges to which academic researchers can contribute. These include: ■ air corridors ■ AA communications ■ 3GPP support for navigation ■ DAA/collision avoidance.

- Air Corridor
 - The design of air corridors follows the overall goal of providing efficient and safe operation of UASs, respecting UVRs and environmental constraints
 - Still, there are numerous open questions, and here we mention a few: ■ Which air corridor concept is the safest one? ■ Which requires the least amount of coordination for collision avoidance? ■ Which provides the highest flexibility to cope with growing UAS traffic? ■ Which guarantees the shortest flight times? ■ Which allows for priority flights? ■ Which can handle the bottleneck of locations where the rates of takeoffs and landings are extremely high (like at a distribution warehouse)? ■ Which is most robust when hazards emerge?
- The most obvious choice for this is AA vehicle-to-vehicle (V2V) communications among the UASs. [The term V2V has been used for approximately 15 years to mean automobile-to-automobile (car-to-car) links. Hence, we discourage its use for airborne vehicles.]
- Standardization efforts for AA communication protocols, use cases, spectrum needs, and security requirements are currently taking place in the AAM community.
- AA communications can be combined with the principles of radar systems to develop dual-function radar communication (DFRC) systems [4] where radar waveforms can be designed to carry information to communicate with other vehicles in the vicinity. The dual use of DFRC for sensing and communications has potential applications in AAM, particularly in the CNS domain.
- AA communications, like air-to-ground (AG) communications, have a history nearly as long as aviation itself. As with terrestrial settings, multiple network topologies (e.g., mesh, relay, star, and so on) can be employed. A primary difference from terrestrial networks arises from high aircraft speeds: AA network topologies can change rapidly. Until air traffic densities become much larger than presently envisioned for AAM applications, AA communications will likely serve a supporting role to primary AG communication links, whereas navigation, and, in particular, surveillance, will rely more on **AA signaling**
- For all three functions (CNS), reliability at the lower layers of the communications protocol stack is critical.
- **AA signaling at the physical and data link layers.**
- When link distances are on the order of altitudes or smaller, the AA channel is very close to a free-space channel, assuming mostly azimuth covering antennas. This is easily deduced from geometry and basic propagation principles. The primary obstacle that affects the AA channel is the Earth surface, and, as link distances increase, reflections from the Earth surface become the next significant channel component in addition to the line-of-sight (LOS) component [5]. For lower aircraft altitudes, the AA link range will decrease since the radio LOS over the curved Earth is proportional to the square root of the altitude. In addition, the probability of obstruction of the AA link increases as the altitude decreases.
- For links between aircraft that are not too close to the ground and are at approximately the same altitude, this $1/d^2$ free-space path loss model allows large link ranges in comparison to terrestrial settings, where, often, a $1/d^3$ or $1/d^4$ attenuation relation holds.
- **this will depend on the link geometry, aircraft altitude, and relative terrain roughness.**
- a well designed simple monopole antenna will have its antenna pattern changed—sometimes drastically—when mounted on an aircraft. Mounting location is also important since an antenna mounted on one side of the fuselage (e.g., the top) will typically suffer significant gain reduction if used to receive a signal coming from the direction of the other side of the fuselage (e.g., the bottom). Thus, multiple antennas distributed across the aircraft may be necessary if connectivity is truly required to be 3D.
- Directional antennas suppress MPCs and can enable larger link ranges, but, in a dynamic AA setting, antenna pointing and tracking can be challenging. For AAM applications, some of the most

challenging cases will be in the near-urban and urban environments. In the near urban case, flights will likely follow established air corridors well above buildings; hence, AA channels may be well modeled by a two-ray channel, in which the terrain cover type will determine the strength and nature of the reflection (specular or diffuse). Work on estimating and validating these air corridor channel models should be done for the appropriate frequency bands.

- Various flight scenarios were assessed with the goal of measuring the propagation conditions in challenging scenarios: e.g., two drones flying one behind the other along building facades at a close distance to the building (see Figure 4), a drone landing in the inner courtyard of a building while another one flies around the building, and two drones flying on a potential collision course toward the corner of a building. Inherently, LOS as well as non-LOS situations have been assessed. The first findings reveal that the channel exhibits a large variety of propagation phenomena encompassing single, double, and triple reflections; diffraction; scattering; shadowing; and combinations. While evaluations are ongoing, the existing results already indicate that the AA channel for urban scenarios in which drones fly below rooftops is at least as complex as car-to-car channels.
- identify critical propagation scenarios like the occurrence of strong reflections potentially arising from large window facades or long-lasting shadowing. Those scenarios are crucial for DAA since they may either interrupt the AA communication link or result in an incorrect reception of messages.
- In release 15, 3GPP conducted a study to investigate the ability of the LTE network to provide connectivity for low altitude UASs. The study defined performance requirements for both command and control as well as for UAS application data. From 3GPP's perspective, both categories are "user-plane" traffic. For command and control, the latency and bit error rate were concluded to be more important than the data rate, which is more crucial than application data, especially with a video use case. Further, channel models and typical scenarios were defined for studying the potential issues and potential solutions to be addressed later in the work item phase. The outcome of the study was documented in the 3GPP technical report [9], which also included field measurement results. In [10], the authors present a related study and findings of UAS operation in cellular networks.
- signal strength measured at the UAS when N neighboring base stations' (BSs') signal strengths are above a threshold. This, together with the height reporting, is used for both interference detection and flying mode detection.
- One is network tracking, where the location information of the UAS is conveyed back to a UAS service supplier.
- The release 18 UAS discussions also include considering further enhancements for uplink beamforming and beam management to further control the interference and to improve the link quality with faster switching of uplink beams toward the BS
- To enable better mobility performance, there are proposals to use and enhance a so-called conditional handover (HO). In a regular HO, the mobile device starts connecting to the target cell immediately upon receiving the HO command. In the case of a conditional HO, the mobile device receives an HO command with a condition, and, when the condition is fulfilled, the device starts connecting to the target cell. Currently, the conditional HO command may include up to eight target cells and up to two conditions per target cell, and the conditions can be signal strength related only.
- These could be used independently or in combination with a signal strength-specific condition.
- However, even when flight trajectories are carefully planned in advance, many conflict situations can arise: ■ High-priority flights may force UASs to change courses, to decelerate, or to stop for a moment and, in turn, the UAS has to react to these changes. ■ Unforeseeable events, like strong wind gusts or mechanical failures, can cause collision courses, which have to be resolved in very

short time. ■ Flight corrections to avoid collisions, e.g., with birds, can result in further collision courses.

- The DAA has two functions: “detect” aims to get awareness about potential collision courses, and “avoid” provides correction maneuvers for all conflicting UASs. “Detect” gets the required information over a communication link, be it AA communication or AG communication, as well as from local sensors, such as cameras, radars, and lidars.
- Further research on the robustness and availability of communication links considering a variety of hazards and appropriate countermeasures is very important.
- The RF signals between a drone and its controller can also be passively monitored for detecting, classifying, localizing, and tracking a drone [15].
- Radars can be used to not only detect and track the drones but also classify them using their measured radar cross section and additional micro-Doppler features
- [2] [4] [9] [10] [11]
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Advancing the Standards for Unmanned Air System Communications, Navigation and Surveillance

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- An analysis of architectures for the two categories of airspace and an implementation technology readiness analysis were performed. These studies produced NASA reports that have been made available in the public domain and have been briefed in previous conferences. We now consider how the products of the study are influencing emerging directions in the aviation standards communities.
- The International Civil Aviation Organization (ICAO) Communications Panel (CP), Working Group I (WG-I) is currently developing a communications network architecture known as the Aeronautical Telecommunications Network with Internet Protocol Services (ATN/IPS).
- However, the work is more and more also considering the emerging class of airspace users known as Remotely Piloted Aircraft Systems (RPAS), which refers to certain UAS classes.
- This paper presents UAS CNS architecture concepts developed under the NASA program that apply to all three of the aforementioned committees

- In the following section, we present an overview of the communication networks, communication data links, navigation and surveillance options considered or developed in our project.
- The International Civil Aviation Organization (ICAO) is therefore developing a new data communications architecture known as the **Aeronautical Telecommunications Network with Internet Protocol Services (ATN/IPS) [1]**.
- These mobility events are known to cause stability issues for standard Internet routing protocols such as the Border Gateway Protocol (BGP) and Open Shortest Path First (OSPF). Second, UAS typically have multiple data link technologies, e.g., satellite links, cellular links, Wi-Fi links, etc., where the links may only be available during certain phases of flight and may even exhibit variable performance characteristics within each flight phase. The coordination of these multiple data links, therefore, becomes challenging from an Internetworking perspective.
- This means that each UAS must be assigned a unique Internet Protocol address or prefix so that routing can direct packets to and from the correct UAS. Since the current global Internet Protocol, version 4 (IPv4) service has run out of addresses; this can only be accommodated by adopting Internet Protocol, version 6 (IPv6). Furthermore, since standard Internet routing services including BGP and OSPF are not equipped to manage large numbers of highly mobile nodes, a new mobility-capable routing service is needed. A nominal mobile routing architecture known as **Asymmetric Extended Route Optimization (AERO) [3]** was developed under our project.
- **incorporate more and more levels of autonomy.**
- Unfortunately, cellular technology is implemented only mostly along the highways and only near populated areas. The cellular signal in remote areas is non-existent or weak. So the UAs using cellular signal will have to follow the highways and not in a straight line between the source and destination. Another problem is that cellular infrastructure is designed and optimized for ground communications. Cellular signals have lobes pointing towards the ground. So the signal reaching skywards is less. However, it is compensated partly by the absence of interference in the skyward direction.
- The UAs using cellular will have to be designed with autonomous navigation so that they can operate autonomously in-between cell towers and sync up with the pilot when they reach the next tower.
- Enhanced Spectral Efficiency, Spiral Approach to Data Link Development, Commercial Non-Aeronautical Satellites
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